

AI-HAZOP-8800

A Guideword-Based Methodology for Identifying and Mitigating Hazards Related to AI Components in Road Vehicles

White Paper 1: Methodology and General Mitigation Measures

Version 0.5 - Draft

Executive summary

This white paper proposes AI-HAZOP-8800, a systematic hazard elicitation and mitigation methodology for AI components used in safety-related road-vehicle functions. The method is based on the intent of HAZOP by applying structured guidewords and questions to AI component aspects: inputs, outputs, internal states, model artefacts, training basis, runtime platform, and downstream interfaces. It is designed for use together with safety engineering activities associated with ISO 26262, ISO 21448, and ISO/PAS 8800:2024. Public ISO information describes ISO/PAS 8800 as a framework for managing AI safety that tailors or extends approaches from ISO 26262 and ISO 21448 and applies to safety-related road-vehicle E/E systems using AI technology.

The methodology focuses on AI insufficiencies and AI-related malfunctioning behavior that can lead to undesired vehicle-level safety behavior. It supports both known hazard discovery and unknown-unknown screening through instability probes, metamorphic testing, uncertainty analysis, OOD detection, field monitoring, and cross-component inconsistency checks.

The paper also proposes a mitigation measure allocation using common approaches: Respecification, Safety Function, and Passive/Operational measures. Respecification changes the AI function or its interface so that the hazardous failure is less likely. Safety functions independently detect, control, or mitigate the hazardous failure. Passive measures reduce exposure or support safe operation through ODD restrictions, maintenance, training, and field monitoring.

The final goal is to produce a comprehensible, defensible and systematic method applicable to AI yet traditional in composition to allow for safety case creation which more easily falls in the scope of traditional safety assessments.

Intended use

- As a checklist and worksheet basis for AI component hazard analysis.
- As a bridge between AI-specific analysis and conventional safety lifecycle artefacts.
- As input to AI Safety Goals, AI Safety Requirements, validation plans, safety cases, and field monitoring plans.
- As a reusable method for perception, fusion, localization, planning, and control AI components.

1. Scope, assumptions, and terminology

1.1 Scope

AI-HAZOP-8800 applies to AI components within safety-related road-vehicle functions. It is especially relevant for components implemented through machine learning, neural networks, statistical models, learned planners or controllers, perception models, prediction models, and AI-assisted decision logic. The method is not limited to perception; it can be applied to camera input processing, segmentation, object detection, tracking, lidar/radar processing, sensor fusion, localization, planning, trajectory generation, and control.

1.2 Core safety question

For every AI component aspect, the analyst asks: Can this input, output, internal state, training assumption, runtime condition, or interface behavior become wrong, late, missing, unstable,

overconfident, biased, corrupted, stale, inconsistent, or otherwise unsafe in a way that may contribute to vehicle-level harm?

1.3 Terminology

Term	Meaning in this white paper
AI component	A software component using AI/ML or learned behavior whose output can influence vehicle-level safety.
AI aspect	An analyzable part of the AI component: input, output, internal state, model artefact, training basis, runtime platform, or interface.
Guideword	A structured prompt used to trigger systematic deviation thinking, e.g., No, More, Less, Wrong, Late, Frozen, Overconfident, Out-of-Distribution (OOD).
Hazardous failure mode	A component-level deviation that can contribute to hazardous vehicle behavior in a given Operational Design Domain (ODD) and scenario.
AI Safety Goal	A high-level safety objective aimed at preventing or controlling AI-related hazardous behavior.
AI Safety Requirement	A concrete requirement allocated to AI design, monitoring, runtime control, validation, or operation.
Respecification	Changing the AI function, its data, model, logic, interface, or performance contract to reduce the failure probability.
Safety Function	Independent or sufficiently separated mechanism that detects, controls, mitigates, or transitions to safe operation.
Passive / Operational measure	Measure that reduces exposure or improves operational control through ODD restriction, maintenance, training, documentation, or field monitoring.
Unknown-unknown candidate	Abnormal behavior pattern not yet mapped to a known failure mode but potentially safety relevant.

2. AI-HAZOP-8800 process overview

The proposed process is intentionally mechanical. The purpose is to avoid relying only on expert intuition and to force repeated variation of component, aspect, ODD, scenario, class, and guideword.

1. Select the next architecture component or next requirement containing components.
2. Determine whether the component is an AI component or contains AI behavior relevant to safety.
3. List all AI component aspects: inputs, outputs, internal states, model artefacts, training basis, runtime platform, and downstream interfaces.
4. Classify the component using the autonomous-driving AI taxonomy in this paper.
5. Select ODD slices and scenario families relevant to the component and output.
6. For each component aspect, apply generic guidewords, class-specific questions, and unknown-unknown probes.
7. For each plausible deviation, formulate the failure mode, hazardous vehicle-level behavior, and potential harm.
8. Estimate risk using MEM-style or GAME/GAMAB-style acceptance reasoning, with clear assumptions.
9. If risk is unacceptable or insufficiently understood, define AI Safety Goals.
10. Allocate AI Safety Requirements and measures: Respecification, Safety Function, Passive/Operational, or a combination.
11. Define tests and evidence: dataset audits, scenario tests, fault injection, runtime monitors, closed-loop simulation, track tests, and field monitoring.
12. Feed new findings and unknown-unknown clusters back into the question catalogue.

2.1 Analysis tuple

The recommended analysis tuple is:

AI component x AI aspect x ODD x Scenario x Class x Guideword/Question -> Failure mode -> Hazardous behavior -> Risk -> AI Safety Goal -> AI Safety Requirements -> Tests/Evidence

2.2 Component aspect checklist

Aspect	What to list	Typical hazard contribution
Inputs	Raw sensors, preprocessed tensors, object lists, tracks, maps, ego pose, route, driver request, V2X/cloud input, actuator feedback.	Missing, degraded, delayed, misordered, biased, corrupted, or OOD input can cause unsafe AI output.
Outputs	Classes, masks, detections, tracks, predictions, fused objects, pose, trajectory, command, brake/accel/steer request.	Wrong, late, unstable, overconfident, stale, or inconsistent output can mislead downstream functions.
Internal states	Recurrent memory, track state, confidence history, object lifecycle, planner state, fallback state, map cache, learned latent state.	State contamination, drift, hysteresis, hanging states, delayed updates, or unsafe mode transitions.
Model artefacts	Architecture, weights, thresholds, calibration parameters, quantization, preprocessing, postprocessing.	Version mismatch, threshold brittleness, quantization error, non-bit-exactness, incompatible preprocessing.
Training basis	Dataset, labels, ODD coverage, scenario coverage, edge cases, class balance, synthetic data, validation split.	Dataset gaps, bias, overfitting, shortcut learning, poor generalization, sim-to-real gap.
Runtime platform	Compute, memory, accelerator, OS, middleware, clocks, health monitors, watchdogs.	Deadline miss, thermal throttling, memory pressure, stale messages, r6esource contention, invalid tensors.
Downstream interface	Interface schema, units, timestamps, validity flags, uncertainty contract, safety assumptions.	Valid-looking but unsafe abstraction, unit/sign errors, missing uncertainty, misuse by planner/control.

3. AI component taxonomy

The taxonomy is used to select class-specific question sets. A component can belong to more than one class; in that case, apply all relevant question sets.

Class	Purpose	Typical inputs	Typical outputs
Camera-Input	Image acquisition and preprocessing	Raw frames, camera health, exposure, intrinsic/extrinsic calibration	Image tensor, metadata, quality indicators
Camera-Segmentation	Pixel/region semantic classification	Image/BEV tensors	Road, lane, curb, pedestrian, cyclist, vehicle, background masks
Camera-ObjectDetection	Detection and classification of objects	Images, ROIs, priors	Bounding boxes, classes, confidence, object counts
Camera-Tracking	Temporal association of detections	Detections, previous tracks, ego motion	Track IDs, velocities, lifecycles, object histories

Lidar	Point-cloud perception	Point clouds, ego motion, calibration	3D objects, free space, obstacle clusters
Radar	Radar object processing	Radar returns, Doppler, ego motion	Objects, range, relative velocity, confidence
Ultrasonic	Short-range obstacle perception	Ultrasonic echoes, vehicle speed	Near-field obstacle presence/distance
SensorFusion	Multi-sensor association and interpretation	Camera/lidar/radar/ultrasonic/map/object data	Fused objects, confidence, uncertainty, free space
Planning-Route	Route and mission planning	Map, destination, ODD limits	Route, lane-level path, stop/pickup points
Planning-Behavior	Tactical decisions	World model, rules, prediction, route	Yield/go/stop/lane-change/overtake decisions
Planning-Trajectory	Motion trajectory generation	Behavior decision, objects, map, ego state	Trajectory, speed profile, constraints
Control-Command	Actuator command generation	Trajectory, ego state, actuator feedback	Steer/brake/accel/torque commands
Control-Dynamics	Vehicle dynamics control	State estimates, model, road condition	Stabilized path following and dynamic behavior
Control-Braking	AI-influenced braking	Objects, trajectory, speed, friction estimate	Brake request, deceleration, stop decision
Control-Acceleration	AI-influenced acceleration	Objects, route, speed target, gap	Throttle/torque/accel request
Localization-HDMap	Map-based localization and semantic map use	HD map, GNSS, perception landmarks	Lane/map relation, semantic priors, ODD/geofence
Localization-PoseEstimation	Ego pose estimation	GNSS, IMU, wheel odometry, lidar/camera localization	Position, heading, covariance, lane-relative pose

4. Generic guideword catalogue

Guideword	Generic meaning	Typical AI examples
No / Missing	Expected signal or output absent.	No object, no mask, no command, no confidence, no internal-state update.
More	Quantity, magnitude, confidence, or effect too high.	Too many detections, excessive braking, over-segmentation, high false positives.
Less	Quantity, magnitude, confidence, or effect too low.	Undercount, weak brake, small object extent, low recall.
Wrong	Incorrect semantic, numeric, spatial, temporal, or logical value.	Wrong class, wrong lane, wrong object velocity, wrong command sign.
Late	Output arrives after safety-relevant deadline.	Delayed pedestrian detection, stale pose, late braking command.
Early	Output/action occurs too soon or prematurely.	Premature lane change, premature braking, early deletion of track.
Intermittent	Output alternates or flickers.	0/1 cyclist count flicker, track lost/reappearing.
Frozen / Stale	Old output reused as current.	Frozen frame, stale object list, unchanged internal state.
Corrupted	Data value invalid or damaged.	NaN, Inf, overflow, negative count, invalid tensor, wrong serialization.
Inverted	Meaning, sign, side, direction, or mapping reversed.	Left/right swap, brake/accel inversion, near/far swap.
Misordered	Order, sequence, lifecycle, or time alignment wrong.	Out-of-order frames, old route segment, wrong track history.
Inconsistent	Contradicts other data or physical plausibility.	Count says zero but box exists; map and perception disagree.
Overgeneralized	Model applies learned pattern too broadly.	Classifies unusual object as known class with high confidence.
Overfitted	Good benchmark behavior but poor deployment generalization.	Fails on new country, route, sensor, lighting, infrastructure.
Biased	Performance differs across ODD slices, object types, locations, or groups.	Poor detection of cargo bikes, children, winter clothing, night cyclists.
Uncertain but confident	High confidence despite ambiguity or unknown input.	High-confidence false negative or false positive.
Certain but unstable	High confidence but changing output.	Alternating class labels with high confidence.

Resource-limited	Runtime resources affect AI quality or timing.	GPU load causes skipped frames or longer inference.
Unmonitored	Failure not detected by safety mechanism.	No freshness, range, confidence, or OOD monitor.
Unsafe fallback	Degradation response creates another hazard.	Sudden stop in traffic, feature turns off without safe transition.
Valid-but-unsafe	Interface value passes checks but is semantically unsafe.	Count alone used without position, velocity, uncertainty, or freshness.

5. Class-specific question index

Camera-Input

- Can the frame be missing, black, frozen, duplicated, partially transmitted, over-compressed, or corrupted?
- Can dirt, mud, rain, snow, insects, fogging, cracks, glare, shadows, or camera housing damage degrade the image?
- Can low sun, tunnel transitions, night, LED flicker, headlights, reflections, wet road, or emergency lights hide or create features?
- Can lens distortion, rolling shutter, vibration, mounting shift, thermal expansion, or calibration drift alter geometry?
- Can frames arrive late, out of order, with wrong timestamp, variable frame rate, or poor synchronization?
- Can crop, resize, color conversion, normalization, compression, ISP tuning, or ROI selection remove safety-relevant content?

Camera-Segmentation

- Can sidewalk, shoulder, grass, construction area, opposite lane, tram lane, or pedestrian zone be marked drivable?
- Can valid road, temporary lane, ramp, narrow passage, or unmarked road be rejected?
- Can curb, edge, barrier, island, pothole, speed bump, or drop-off be missed or shifted?
- Can small/thin objects such as poles, cones, chains, strollers, wheelchairs, pets, debris, or open doors disappear?
- Can temporal mask flicker or lag when scene changes?
- Can low-confidence segmentation be consumed as valid?

Camera-ObjectDetection

- Can pedestrians, cyclists, motorcycles, scooters, animals, road workers, trailers, low obstacles, cones, or barriers be missed?
- Can shadows, signs, reflections, vegetation, ads, markings, or rain artefacts become ghost objects?
- Can pedestrian/cyclist/motorcycle/scooter/vehicle/static/moving classes be confused?
- Can bounding box position, size, distance, extent, or yaw be wrong?

- Can occlusion, truncation, crowding, groups, object emergence, scale, or distance cause missed or duplicated objects?
- Are confidence scores calibrated and meaningful per class and ODD?

Camera-Tracking

- Can real object tracks fail to start, be deleted too early, persist after disappearance, or duplicate?
- Can IDs swap between nearby objects or crossing objects?
- Can velocity, acceleration, heading, intent, or path history be wrong?
- Does the tracker bridge occlusion safely or hallucinate?
- Can stale state dominate new evidence or recover too slowly?
- Is uncertainty propagated to prediction and planning?

Lidar

- Can point clouds be sparse, noisy, missing, duplicated, saturated, reflected, or corrupted?
- Can rain, fog, snow, dust, spray, black surfaces, glass, mirrors, retroreflectors, or wet road affect returns?
- Can calibration, mounting, vibration, ego-motion compensation, scan pattern, or time sync fail?
- Can low, thin, overhanging, transparent, absorbent, or irregular objects be missed?
- Can ground removal suppress curbs, slopes, ramps, speed bumps, potholes, or obstacles?
- Can clustering merge, split, ghost, or misdimension objects?

Radar

- Can stationary, slow, crossing, low-RCS, or lateral-moving objects be missed?
- Can multipath, guardrails, metal structures, tunnels, parked vehicles, or other radars create ghosts?
- Can Doppler ambiguity or ego-motion compensation cause wrong velocity?
- Can range, azimuth, elevation, or object extent be wrong?
- Can weather or clutter mask targets?
- Are radar frames synchronized with camera/lidar?

Ultrasonic

- Can poles, curbs, children, pets, bicycle wheels, low walls, chains, glass, soft material, or angled surfaces be missed?
- Can rain, snow, puddles, grass, exhaust, other sensors, or reflections create false echoes?
- Are blind zones known around bumpers and corners?
- Is vehicle speed compatible with update rate?
- Can dirt, ice, bumper deformation, sensor paint, or misalignment degrade readings?

SensorFusion

- What if camera, lidar, radar, ultrasonic, and map disagree?
- Can ghost evidence from different sources be fused into a false object?
- Can real object be rejected because one sensor disagrees?
- Can calibration drift cause object displacement?
- Can asynchronous frames combine objects from different times?

- Can disagreement be converted into overconfidence rather than uncertainty?

Planning-Route

- Can the route leave the approved ODD or use prohibited/closed/unsuitable roads?
- Can HD map, navigation map, live restrictions, traffic rules, or roadworks data be outdated?
- Can wrong destination, stop point, pickup/drop-off zone, or lane-level goal be selected?
- Can route update or reroute be late?
- Can route force unsafe/illegal maneuver?

Planning-Behavior

- Can right-of-way, yield, stop, crosswalk, merge, roundabout, or emergency vehicle handling be wrong?
- Can pedestrian/cyclist/vehicle intent be mispredicted?
- Can gap acceptance be too small or overly timid?
- Can behavior oscillate between yield/go/stop/lane-change?
- Can roadworks, police direction, blocked lane, hand gestures, or unprotected turns be mishandled?

Planning-Trajectory

- Can trajectory intersect an object, predicted path, curb, barrier, oncoming lane, sidewalk, or low obstacle?
- Is the trajectory feasible for vehicle limits, friction, slope, payload, braking distance, and actuator limits?
- Can comfort smoothing delay emergency action?
- Are speed, acceleration, jerk, lateral acceleration, steering angle, and stopping distance respected?
- Are object, prediction, and localization uncertainties included?

Control-Command

- Can steering/throttle/brake command have wrong magnitude, sign, unit, coordinate frame, or scaling?
- Can commands be late, repeated, skipped, reordered, or generated from stale trajectory?
- Can commands exceed actuator limits or be clipped silently?
- Can AI command conflict with safety controller, driver input, ESC, ABS, or fallback controller?
- Can rounding, fixed-point conversion, accelerator differences, or non-bit-exactness materially change output?

Control-Dynamics

- Is the learned or model-based controller valid for payload, tires, friction, slope, wind, battery, brake temperature?
- Can model mismatch, overfitting, oscillation, overshoot, instability, or delayed correction occur?
- Is low-friction road handled: ice, wet leaves, gravel, snow, mud?
- Are yaw rate, slip angle, speed, acceleration, and wheel speed reliable?

Control-Braking

- Can braking be absent, late, insufficient, excessive, or unintended?

- Is stopping distance computed with friction, slope, load, tire state, delay?
- Does AI braking conflict with AEB, ABS, ESC, or regenerative braking?
- Can jerk limits or comfort filters delay safety braking?
- Could false braking create secondary risk?

Control-Acceleration

- Can acceleration occur when stop/yield/brake is required?
- Can acceleration be too high, too low, delayed, oscillatory, or sustained?
- Is low-speed creep safe near pedestrians, crossings, parking, stations?
- Can acceleration reduce safety margin during cut-in or merge?
- Can throttle/brake sign, torque request, or regen blending be wrong?

Localization-HDMap

- Can lane geometry, signs, signals, speed limits, crosswalks, curbs, stop lines, construction zones be outdated or wrong?
- Are ODD boundaries, geofences, no-go zones, depot/station areas represented?
- Can vehicle match to wrong lane, road level, parallel road, tunnel, bridge, or parking lane?
- Can map update be delayed, partial, corrupted, incompatible, or unverified?
- Can wrong priority, lane direction, turn restriction, or pedestrian area be encoded?

Localization-PoseEstimation

- Can pose drift, jump, bias, or covariance underestimation occur?
- Can yaw, pitch, roll, heading, or lane-relative pose be wrong?
- What happens with GNSS outage, multipath, tunnel, urban canyon, wheel slip, IMU drift, or localization failure?
- Is pose timestamp aligned with perception/planning/control?
- Can pose be globally plausible but wrong relative to lane/map?

6. Unknown-unknown probe catalogue

Unknown-unknown probes look for abnormal behavior patterns that are not yet assigned to a known use case or known failure mode. When a probe reveals unexplained abnormality, create an Unknown-Unknown Candidate Hazard, cluster similar cases, investigate causality, apply temporary conservative mitigation, and update the AI-HAZOP catalogue.

Probe family	Questions / hints
Output instability	- Does the output change significantly for small input changes? - Does it oscillate although the scene is static? - Does confidence remain high while output changes rapidly? - Does output behavior depend on previous unrelated scenes? - Does behavior differ after long runtime vs startup?
Metamorphic consistency	- Do moderate brightness, contrast, noise, resolution, or frame-rate changes preserve the semantic output? - Does object distance or position change output gradually rather than catastrophically? - Does partial occlusion degrade gracefully? - Are equivalent sensor representations handled equivalently?
Confidence and uncertainty	- Is the component highly confident on unfamiliar inputs? - Does uncertainty increase under occlusion, blur, glare, rain, partial information, or ODD boundary? - Does uncertainty correlate with actual error? - Are low-confidence outputs still treated as reliable?
OOD / novelty	- Can the component detect that the input is unlike training data? - Does it output normal-looking results for abnormal inputs? - Does it detect rare combinations rather than only rare features? - Do latent embeddings reveal clusters of unexplained errors?
Temporal	- Does performance degrade during long sequences? - Does one bad frame poison later outputs? - Does error grow slowly without immediate visible failure? - Does output latency vary with scene complexity?
Boundary / transition	- What happens at ODD boundaries, tunnel transitions, weather changes, disappearing markings, speed thresholds, confidence thresholds, fallback entry/exit? - Does threshold chatter occur?
Cross-component	- Do components disagree in unexplained ways? - Does fusion convert disagreement into overconfidence? - Does downstream logic amplify small upstream errors or suppress uncertainty? - Does fallback rely on the same failed AI evidence?
Runtime	- Does quality degrade under CPU/GPU load, memory pressure, thermal throttling, compiler/runtime changes, or quantization? - Are NaN, Inf, invalid tensors, empty arrays, stale valid-looking outputs detected?

Dataset blind spots	- Are failures clustered by route, season, country, road type, lighting, object type, or sensor configuration? - Does high average performance hide poor slice performance? - Are rare combinations and safety-critical edge cases underweighted?
Shortcut learning	- Does classification depend on background, lane markings, usual object location, color, texture, road signs, simulation artefacts, or co-occurring objects? - Does object-only performance collapse? - Does attention focus on irrelevant context?
Stress / fuzz	- What happens under physically plausible perturbations, combined mild degradations, unusual object shapes, rare legal behavior, and valid inputs near limits?
Field monitoring	- Are disengagements, near misses, low-confidence clusters, fallback activations, and no-fault-found events accumulating without mapped cause?

7. Risk assessment and acceptance reasoning

AI-HAZOP-8800 does not prescribe one universal numeric target. It supports MEM-style individual risk reasoning and GAME/GAMAB-style comparative reasoning. The selected criterion and allocation must be justified by the project safety concept, ODD, function, regulatory context, and reference system.

7.1 Illustrative risk model

A practical worksheet formula is:

$$R = E \times PF \times PND \times PNM \times S$$

Symbol	Meaning	Example influence of measures
E	Exposure to the scenario/function per person-year.	Reduced by ODD restriction, speed restriction, operating hours, geofence, route design.
PF	Probability that the AI-related failure mode occurs in that scenario.	Reduced by better data, model, architecture, preprocessing, postprocessing, calibration, validation.
PND	Probability that the failure is not detected.	Reduced by monitors, consistency checks, OOD detection, uncertainty calibration, freshness checks.
PNM	Probability that no mitigation prevents harm after failure.	Reduced by fallback, conservative behavior, braking, speed reduction, safety envelope, independent safety controller.
S	Fatality-equivalent severity.	Reduced by speed limitation, physical separation, operational design, safe stop, external protection.

7.2 MEM-style reasoning

MEM is based on the idea that a new technical system shall not significantly increase a person's minimum endogenous mortality. Public PEGASUS material cites minimum natural mortality around 6×10^{-5} per person-year and discusses allocations such as MEM/20 for a new technical risk. Some automated-driving safety literature discusses values around 2.5×10^{-6} per person-year depending on mortality-rate assumptions. In a project, the selected target must be explicitly allocated and justified; this paper often uses 10^{-5} only as an illustrative threshold.

7.3 GAME/GAMAB-style reasoning

GAME/GAMAB means globally at least equivalent / globally at least as good. The residual safety level of the new system shall be at least as good as the selected reference system for a comparable function, ODD, and service. GAME is not one fixed numerical threshold. It requires careful definition of reference, exposure, severity distribution, scenario mix, and evidence that the new system does not make safety worse.

8. Mitigation taxonomy

Mitigation class	Definition	Primary risk factor reduced	Examples
Respecification	Change the AI function, data, model, algorithm, interface, or performance contract so the hazardous failure is less likely or less safety-relevant.	PF, sometimes PNM	Improve recall, add VRU super-class, modify thresholds, add uncertainty output, change INT count to richer safety contract, retrain with ODD slices, remove shortcut learning.
Safety Function	Separated or sufficiently independent monitor/control mechanism that detects the failure, prevents propagation, or transitions to safe operation.	PND and PNM	Freshness watchdog, timestamp monitor, OOD detector, cross-sensor consistency, safety envelope, independent emergency braking, fallback to minimal risk condition.
Passive / Operational	Operational controls and supporting processes that reduce exposure or maintain assumptions.	E, sometimes PF or S	ODD restriction, route restriction, weather restriction, speed limit, maintenance, calibration process, camera cleaning, operator training, field monitoring.
Combined measure	Planned combination of the above where a single measure is insufficient.	PF, PND, PNM, E, S	Improve model, monitor uncertainty, restrict unsupported ODD, reduce speed, collect field data, update safety case.

9. General measures per discovered potential hazard

Hazard family	Recommended general measures
Missing / false negative output	R: improve training data and recall; respecify conservative absence semantics. SF: absence-of-evidence monitor, cross-sensor confirmation, conflict-zone caution. P: restrict ODD where visibility/detection is unsupported.
Undercount / merged objects	R: group-aware model, instance separation, NMS tuning, extended-object representation. SF: uncertain group treated as extended VRU/obstacle area. P: speed limits in dense VRU areas.
Overcount / false positive	R: duplicate suppression, hard-negative mining, better postprocessing. SF: require spatial conflict before harsh maneuver; comfort/safety arbitration. P: monitor nuisance braking rate.
Wrong class	R: confusion-matrix-driven retraining, VRU super-class, taxonomy change. SF: uncertain VRU classification triggers conservative behavior. P: restrict unsupported object types/regions.
Late output	R: optimize model, reduce computational load, define latency budget. SF: latency watchdog, stale-data invalidation, speed reduction when timing margin is insufficient. P: speed/ODD restriction.
Frozen / stale	R: architecture avoids stale reuse. SF: timestamp, heartbeat, frozen-frame, sequence counter, watchdog. P: maintenance and logging review.
Flickering / instability	R: temporal smoothing with safety-preserving semantics, hysteresis, stable thresholds. SF: instability flag, conservative mode on oscillation. P: field monitoring for unstable routes/slices.
Corruption / invalid value	R: robust interface schema and data contracts. SF: type/range/NaN/Inf/overflow checks and safe rejection. P: software configuration management.
Misordered / unsynchronized	R: deterministic dataflow and explicit time-base design. SF: monotonic timestamp and synchronization monitor. P: integration test procedures.
Inconsistent outputs	R: consistent postprocessing and shared object model. SF: count/object/track/fusion plausibility checks. P: safety case review of interface assumptions.
Overconfident wrong output	R: confidence calibration, OOD-aware training, uncertainty outputs. SF: OOD detector and independent evidence requirement. P: field mining of high-confidence errors.
Threshold sensitivity	R: confidence bands, graceful degradation, robust threshold selection. SF: near-threshold candidates propagated as uncertain. P: validation at boundaries.
Shortcut learning	R: counterfactual training/testing, context swapping, object-only validation, saliency checks, hard-negative mining. SF: OOD/context-shift monitor and multi-evidence consistency. P: ODD restriction and dataset maintenance.
Dataset / ODD gap	R: ODD-slice dataset requirements and balanced validation. SF: ODD validity monitor and conservative mode in unsupported slices. P: operational restriction and field data collection.

Multi-sensor duplicate / association error	R: calibration-aware association, overlap-zone logic. SF: duplicate monitor, calibration monitor. P: calibration maintenance.
Correct count but wrong spatial data	R: output contract requires spatial validity, not count alone. SF: uncertainty envelope, box/track validity checks. P: interface review and developer training.
Saturation / max limit	R: interface encodes AT_LEAST, UNKNOWN, INVALID, not only integer. SF: saturation handling as crowd/uncertain obstacle. P: restrict crowd scenarios until validated.
Valid-but-unsafe abstraction	R: replace unsafe simple output with safety contract including freshness, confidence, uncertainty, status, associated objects. SF: downstream misuse monitor. P: architecture review.
Resource-dependent degradation	R: worst-case execution-time and resource design. SF: resource monitor, thermal/load monitor, degraded mode. P: platform qualification and maintenance.
Unsafe fallback	R: respecify fallback strategy and interface. SF: controlled transition to minimal risk condition. P: operator procedure and ODD restriction.
Unknown-unknown candidate	R: update model/data/spec after investigation. SF: temporary conservative monitor or fallback. P: field monitoring, issue triage, catalogue update.

10. Safety-oriented AI output contracts

A repeated finding is that many AI outputs are unsafe when represented as simple values. For example, `N_cyclists_detected : INT` is not sufficient as a safety-relevant contract because it hides uncertainty, spatial validity, freshness, confidence, saturation, and association to object data.

10.1 Recommended contract pattern

For safety-relevant AI outputs, use a structured contract with at least:

- Value and validity status: EXACT, AT_LEAST, UNKNOWN, INVALID.
- Freshness and timestamp.
- Confidence and calibrated uncertainty.
- ODD validity or novelty status.
- Instability flag or recent output variation.
- Associated objects, spatial data, and track references where relevant.
- Saturation indication and lower/upper bounds.
- Explicit safe-consumption rules for downstream components.

Example for cyclist count:

```
CyclistPerceptionSummary {
    count_exact optional;
    count_lower_bound;
    count_upper_bound optional;
    count_status; confidence;
    freshness_ms;
    uncertainty_level;
    instability_flag;
    ODD_status;
    associated_object_ids;
}
```

11. Evidence and test catalogue

Evidence type	Purpose	Typical test activities
Dataset audit	Show that training/validation data support the declared ODD and scenarios.	Coverage matrix, class balance, slice performance, label quality review, synthetic/real split, independence checks.
Scenario-based validation	Show behavior in nominal, critical, edge, and boundary scenarios.	Closed-loop simulation, track tests, replay tests, corner cases, rare object configurations.
Metamorphic testing	Find unknown brittleness without needing perfect ground truth.	Brightness, contrast, noise, occlusion, scale, resolution, frame-rate, background changes.
Fault injection	Demonstrate monitors and fallbacks.	Missing frames, stale outputs, timestamp disorder, corrupt values, invalid tensors, sensor degradation.
Uncertainty/OOD testing	Show detection of situations outside reliable knowledge.	OOD score validation, confidence calibration, high-confidence false-negative mining, novelty clustering.
Runtime verification	Show timing and platform behavior.	WCET, load/thermal tests, memory pressure, quantization, non-bit-exactness, accelerator comparison.
Cross-component consistency	Show that contradictions are detected and safely handled.	Count vs boxes, perception vs map, camera vs lidar/radar, planner vs uncertainty, fallback independence.
Safety case traceability	Connect hazards to goals, requirements, tests, evidence, and residual risk.	Trace matrix, argumentation, assumptions, limitations, open issues.
Field monitoring	Detect residual unknown-unknowns after deployment.	Disengagement analysis, near-miss mining, fallback trends, route clusters, no-fault-found investigations.

12. AI-HAZOP-8800 worksheet template

Field	Content to enter
Hazard ID	Unique identifier.
Component	AI component under analysis.
Class	Taxonomy class or classes.
Aspect	Input, output, state, training basis, runtime platform, interface.
ODD	ODD slice including road type, speed, weather, lighting, geography, traffic density.
Scenario	Concrete operating scenario.
Guideword/question	Question that triggered the failure mode.
Failure mode	Component-level deviation.
Hazardous behavior	Vehicle-level unsafe behavior.
Potential harm	Who/what may be harmed and how severely.
Initial risk	E, PF, PND, PNM, S, R.
Acceptance criterion	MEM target or GAME reference and rationale.
Safety decision	Accept, improve, restrict, or investigate.
AI Safety Goal	High-level safety goal.
Measures	Respecification, Safety Function, Passive/Operational.
Residual risk	Recomputed risk with rationale.
Evidence	Tests, reviews, monitors, field data.
Open assumptions	Unverified assumptions and required follow-up.

13. Lifecycle integration

AI-HAZOP-8800 is not a one-time workshop. It should be repeated at architecture definition, AI component specification, dataset definition, model selection, integration, validation, release, update, and field monitoring. Unknown-unknown findings should update the catalogue and feed new scenario requirements.

- Concept phase: identify AI components, ODDs, scenarios, preliminary hazards, initial AI Safety Goals.
- Development phase: allocate AI Safety Requirements to data, model, interface, monitor, and fallback design.

- Verification phase: perform class-specific tests, fault injection, timing tests, metamorphic tests, uncertainty/OOD tests.
- Validation phase: evaluate closed-loop vehicle behavior and compare residual risk against MEM/GAME criteria.
- Release phase: document limitations, ODD restrictions, safety monitors, fallback strategy, and residual assumptions.
- Operation phase: monitor field behavior, near misses, disengagements, no-fault-found events, and ODD drift.
- Update phase: re-run AI-HAZOP for changed model, data, thresholds, sensors, runtime, ODD, or interface.

14. Mini-example: cyclist count output

The method was tested conceptually on an object detection/classification component that outputs `N_cyclists_detected` : INT. Dominant unacceptable initial hazards were: zero count when cyclist exists, undercounting groups, wrong class, late update, overconfident wrong zero, shortcut learning, ODD/data gaps, and merged cyclists. The strongest mitigations were not passive measures alone, but combined respecification and safety functions.

Example 1 – False Negative Cyclist Detection

System Context

Component	Camera Object Detection
Output	<code>N_cyclists_detected</code> : INT
ODD	University Campus Shuttle
Scenario	Cyclist behind parked van
Guideword	NO
Hazard H-001	
Cyclist exists but detector reports <code>N_cyclists_detected</code> = 0. Vehicle assumes crossing is clear and may fail to yield.	

Initial Risk Assessment

Exposure E	0.03
Failure Probability PF	1E-3
Not Detected PND	0.8
Mitigation Failure PNM	0.6
Severity S	1
Risk R	1.44E-5

STATUS: ABOVE MEM TARGET (NOT ACCEPTABLE)

Mitigation Package

R1 – Respecification
Improve cyclist recall through occluded cyclist datasets, cargo bikes, campus-specific scenes, hard-negative mining, confidence calibration and recall targets. Purpose: reduce PF.
SF1 – Absence-of-Evidence Monitor
CyclistCount=0 is not automatically interpreted as cyclist absence. Monitor evaluates visibility, occlusion, sensor health, confidence and ODD validity. If observability is insufficient, output becomes UNKNOWN instead of NO CYCLISTS. Purpose: reduce PND.
SF2 – Conflict-Zone Caution
Approaching cyclist crossings automatically triggers reduced speed, increased braking readiness and conservative acceleration. Operates independently of perception output. Purpose: reduce PNM and severity.
P1 – ODD Restriction
Operation prohibited in unsupported conditions such as dense fog. Purpose: reduce exposure E.

Residual Risk Assessment

Exposure E	0.025
Failure Probability PF	5E-4
Not Detected PND	0.4
Mitigation Failure PNM	0.4
Severity S	1
Residual Risk R	2.0E-6

STATUS: ACCEPTABLE

Example 2 – Cyclist Group Merged Into One Object

Component	Camera Object Detection
Output	N_cyclists_detected : INT
ODD	University Campus Shuttle
Scenario	Two cyclists side-by-side
Guideword	LESS
Hazard H-002	
Two cyclists exist, but post-processing/NMS suppresses one detection. Planner underestimates occupied cyclist space and may generate unsafe trajectory.	

Root Cause Analysis and Initial Risk Assessment

The neural network correctly detects two cyclists. However, Non-Maximum Suppression (NMS) treats overlapping detections as a single object. This demonstrates that the hazard originates from post-processing rather than the AI model itself.

Exposure E	0.02
Failure Probability PF	0.002
Not Detected PND	0.7
Mitigation Failure PNM	0.5
Severity S	1
Risk R	1.4E-5

STATUS: ABOVE MEM TARGET (NOT ACCEPTABLE)

Mitigation Package

R1 – NMS Tuning
Use cyclist-specific IoU threshold (e.g. 0.75 instead of 0.5). Reduces suppression of adjacent cyclists. Purpose: reduce PF.
R2 – Group-Aware Model
Introduce CyclistGroup representation. Instead of forcing exact counts, the system may report an uncertain cyclist cluster with occupancy width and confidence. Purpose: reduce PF.

SF1 – Extended Object Safety Envelope
Planner treats uncertain cyclist groups as enlarged occupancy regions. Safe trajectory generated around group rather than individual cyclist. Purpose: reduce PNM.

SF2 – Count Consistency Monitor
Cross-check cyclist count against boxes, tracks and fusion outputs. Inconsistency triggers conservative behavior. Purpose: reduce PND.

P1 – Dense VRU Speed Limitation
Campus center speed limited to 10 km/h. Purpose: reduce severity.

Residual Risk Assessment

Exposure E	0.02
Failure Probability PF	8E-4
Not Detected PND	0.4
Mitigation Failure PNM	0.4
Severity S	1
Residual Risk R	2.56E-6

STATUS: ACCEPTABLE

15. References

- ISO/PAS 8800:2024 Road vehicles - Safety and artificial intelligence. Public description: framework for managing AI safety that tailors or extends ISO 26262 and ISO 21448 approaches.